



Sustainable Supply Chain Finance Readiness in Emerging-Economy Manufacturing: A Hybrid Multi-Criteria Decision-Making Approach

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Abstract: Supply chain finance can facilitate access to working capital by leveraging the creditworthiness of focal buyers to improve financing conditions for suppliers, an advantage that is particularly relevant in emerging economies characterized by constrained access to credit. However, existing supply chain finance readiness assessments remain predominantly focused on economic and financial considerations and have largely been developed for mature receivables-financing environments, thereby providing insufficient consideration of the sustainability and organizational conditions encountered in emerging-economy manufacturing. To address this limitation, a hybrid multi-criteria decision-making framework was developed to assess sustainable supply chain finance readiness based on 21 sub-criteria grouped into six sustainability and organizational criteria. The Criteria Importance Through Intercriteria Correlation (CRITIC) method was applied to obtain data-driven sub-criteria weights from assessments provided by 11 industry participants, thereby reducing reliance on explicitly elicited subjective preference weights. Fuzzy Technique for Order Preference by Similarity to Ideal Solution (fuzzy TOPSIS) and grey relational analysis were subsequently employed to prioritize the six criteria. Consistent ranking patterns were obtained using both methods. Social factors were ranked highest, with scores of 0.566 under fuzzy TOPSIS and 0.695 under the grey relational analysis, whereas organizational policies received the lowest scores, at 0.236 and 0.454, respectively. The convergence of the two ranking approaches indicates that sustainable supply chain finance readiness in emerging-economy manufacturing is shaped not only by firm-level financial and operational capacity but also by broader relational and institutional conditions. Accordingly, greater sustainable supply chain finance readiness may be supported by strengthening inter-organizational coordination, stakeholder relationships, and institutional support while simultaneously developing firm-level capabilities. The proposed framework provides a structured basis for identifying priority dimensions of sustainable supply chain finance readiness and may support more context-sensitive financing and sustainability strategies in emerging-economy manufacturing supply chains.

Keywords: Sustainable supply chain finance; Multi-criteria decision-making; Criteria Importance through Intercriteria Correlation; Fuzzy Technique for Order Preference by Similarity to Ideal Solution; Grey relational analysis; Emerging economy

1 Introduction

The trade finance gap globally was estimated at about USD 2.5 trillion, 10% of the total value of world trade, in 2025, and companies from emerging economies continue to be the worst-hit [1]. Supply chain finance is the major solution for the closure of this gap. By aligning the flow of money through the trading partners, supply chain finance facilitates the ability of the supplier to get loans depending on the credit status of the buyer as well as on transaction information rather than the balance sheet only [2]. The reverse factoring method, which has been extensively studied, finances the invoices based on their discount that depends on the credit status of the buyer [3]. The use of such

systems has largely been driven by funding pressures and lengthy fulfillment periods, and it leads to reductions in supply chain costs [4].

Supply chain finance has now been considered less as a form of financial transaction and more as a capacity based on the coordination of financial and informational flows [5], with digital platforms having taken this coordination to new levels of sophistication beyond financing [6]. Sustainable development has become part of this framework. Sustainable supply chain finance acknowledges environmental and social impacts as goals of the financing process rather than as limiting factors, and sustainable supply chain finance has been described as an “ecosystem” where financial and sustainability information create common value [7]. The very same review is quite clear on the fact that the criteria being used in assessments are still primarily economic in nature.

Existing studies on supply chain finance have predominantly focused on developed economies, where receivables-financing systems and supporting institutional frameworks are relatively mature. These studies commonly assume that buyers have transparent and reliable credit ratings, receivables can be legally transferred without significant difficulty, and financial institutions are willing to incorporate transaction-level information into lending decisions. However, these assumptions may not hold in emerging economies, where financing decisions may depend more heavily on long-term business relationships, trust, buyer cooperation, and institutional support. Evidence from Ghana indicates that buyer cooperation and the digitalization of supply chain operations facilitate supply chain finance adoption among small firms [8]. Similarly, improved access to external financing has been positively associated with the productivity of small and medium-sized enterprises in emerging economies [9], while exporters are often required to comply with environmental and social standards imposed by international buyers [10]. Therefore, supply chain finance criteria established in developed economies cannot be directly generalized to emerging-economy contexts without empirical assessment.

Assessing readiness for sustainable supply chain finance requires the simultaneous consideration of multiple economic, environmental, social, operational, managerial, and organizational factors. Consequently, the readiness assessment represents a multi-criteria decision-making problem. Previous studies have modeled sustainable supply chain finance under uncertainty by incorporating economic, environmental, and social criteria [11] and have evaluated multiple criteria using integrated methods such as the best–worst method, Technique for Order Preference by Similarity to Ideal Solution (TOPSIS), and TOMada de Decisão Interativa e Multicritério (TODIM; Interactive and Multicriteria Decision Making) [12]. However, preference-based weighting methods depend heavily on subjective expert judgments and may become increasingly difficult to apply consistently as the number of criteria increases [13]. Moreover, many previous studies have relied on a single ranking method, providing limited evidence regarding the robustness of the resulting priorities. These limitations highlight the need for an objective weighting procedure combined with multiple complementary ranking methods.

Accordingly, this study addresses the following research questions:

RQ1: What criteria and sub-criteria determine readiness for sustainable supply chain finance in the manufacturing industry of an emerging economy?

RQ2: How can these criteria and sub-criteria be weighted from assessment data and prioritized to identify the areas requiring the greatest improvement?

To answer these questions, the study pursues the following objectives:

- To identify and structure the key criteria and sub-criteria influencing sustainable supply chain finance readiness;
- To prioritize them and identify the areas that require the greatest improvement to strengthen sustainable supply chain finance readiness in the manufacturing industry of an emerging economy.

To achieve these objectives, the study develops an assessment framework comprising six main criteria and 21 sub-criteria derived from the supply chain finance, sustainable supply chain, and sustainable supply chain finance literature. The framework extends the traditional triple-bottom-line dimensions by incorporating operational capacity, product and service management, and organizational policies. These additional dimensions are necessary because the triple bottom line primarily defines desired sustainability outcomes but provides limited guidance regarding the organizational capabilities required to achieve financial viability and operational readiness. Data collected from 11 industry participants are analyzed using the Criteria Importance Through Intercriteria Correlation (CRITIC) method [14], fuzzy TOPSIS, and grey relational analysis. CRITIC is selected because it derives sub-criterion weights from the rating data by considering both the variability of criterion performance and the degree of conflict among criteria, thereby reducing reliance on subjective preference-based weighting. Fuzzy TOPSIS is employed to manage ambiguity in expert assessments and prioritize criteria according to their relative distances from ideal and negative-ideal conditions. Grey relational analysis provides a complementary ranking based on the relational closeness of each criterion to the desired reference condition. Using two ranking methods also enables cross-method comparison and strengthens the robustness of the resulting priorities. The remainder of the study is organized as follows. Section 2 reviews supply chain finance, sustainable development, and the multi-criteria methods applied in this study, and the criteria used to measure supply chain finance. Section 3 presents the proposed framework and Section 4 describes the results and discussion. Finally, Section 5 concludes the study by summarizing the key results

and outlining the limitations and future research directions.

2 Literature Review

This section gives an overview of supply chain finance, sustainable development, multi-criteria decision-making approaches and discusses a group of measurements that influence sustainable supply chain finance.

2.1 Supply Chain Finance

Supply chain finance can be described as the management of flows of money between trading partners in a manner that enhances the liquidity and working capital efficiency [2]. The key characteristic of supply chain finance is that the decision on whether or not to extend credit is made by taking into consideration the relationship of the parties rather than just relying on the balance sheet of the borrowing company, such that a supplier is able to access credit on the basis of its buyer's credit rating as well as transaction information. A review of literature reveals that the area has advanced along two distinct but related fronts [15, 16]. There have been enough supply chain finance instruments developed and discussed in literature. One of the mechanisms that has been analyzed to the largest extent is reverse factoring, which involves financing of the invoices from an approved list by the financial institution, based on the credit standing of the buyer. The success of such financing depends on the balance between the discount and payment periods offered by the buyer [3]. Factoring may become a solution to increase access to funding in case of reliable receivable transfer [17]. The implementation of supply chain finance is usually motivated by financial necessity and long order fulfillment time; however, supply chain finance could help in the reduction of supply chain cost [4].

The latest academic research has started to consider supply chain finance as an organizational competence and not merely as a finance-related phenomenon. The effective functioning of supply chain finance relies on coordination of the financial and informational flows, where the information exchange capabilities influence the results of financing [6, 7]. The use of digital platforms facilitates coordination [6], whereas blockchain technologies can enhance the capability of supply chain finance to lower the cost of capital [18]. Likewise, digitalization improves the role of supply chain finance in sustainable development [19]. Supply chain finance is also able to enhance operational resilience in cases of disruption [20]. The concept of sustainable supply chain financing can be seen as the farthest development on this path, in the sense that environmental and social results are considered goals of the financing decision rather than its constraints. It is described as a system, where financial tools and sustainability data are used in tandem to create joint value from economic, environmental, and social perspectives. However, this area remains largely underdeveloped, and its evaluation criteria are still mainly economic [7]. Empirical evidence shows that environmental, social, and governance practices partially moderate the connection between supply chain finance and financial performance [21], thus implying that sustainability practice acts as a medium through which supply chain finance delivers its benefits.

However, there are two gaps in supply chain finance literature. The existing criteria for supply chain finance readiness have not been able to cover the development process of supply chain finance. Most of the literature focuses on economies where the receivables system is well developed. Examples from Bangladesh indicate a lack of external finance availability [9] and the benefits of supply chain finance as far as risk reduction are concerned [22].

2.2 Sustainable Development

The concept of sustainable development refers to the achievement of happiness and financial stability while at the same time maintaining natural resources for future generations' needs [23]. Nonetheless, the process of sustainable development in each nation depends on various factors such as the economy, environment, capacity to sustain, and culture [24]. The Sustainable Development Goals have provided a common framework, and their alignment with environmental, social, and governance considerations has ensured that sustainability goals are integrated into decision-making processes [25]. In the organizational context, the triple bottom line is an approach to sustainable development based on the analysis of performance in economic, environmental, and social terms [26]. Combining all three aspects into the organization's strategy can enhance competitiveness and make sustainability a key element of organizational activities [27]. However, the successful application of the triple bottom line needs collaboration among partners of the supply chain and proper systems of internal control of sustainable performance [28, 29]. As a result, there is an increasing body of literature dedicated to the use of quantitative approaches to analyze sustainable supply chains [30], the transformation of the triple bottom line concept into the measuring instrument [27], and financial tools of sustainability [31].

The context of the emerging economy gives particular relevance to the proposed model. From the ready-made garment industry, it is found that there is substantial progress made on the environmental side but still considerable challenges in balancing economic performance and social considerations, especially for smaller firms and subcontractors [10]. As sustainability in this industry is determined mostly by customer agreements rather than

by individual business decision-making, social and relational performance measures in supply chains might be more significant than strict economic performance measures.

2.3 Multi-Criteria Decision-Making

Evaluation of supply chain finance readiness is a complex process with many related criteria and vague expert opinions; thus, multi-criteria decision-making can be applied to such situations [32]. In this study, this process is divided into two phases: criterion weighting and alternative ranking. As for weighting techniques, subjective methods such as the analytic hierarchy process and the best-worst method are based on pairwise comparisons and may find it hard to sustain consistency as the number of criteria grows. However, objective methods calculate weight directly from evaluation data [13]. The CRITIC method is among those and is popular to use due to consideration of both criteria variability and correlation between criteria, thus reducing the influence of redundant information [14]. Though some improvements have been suggested to cope with linearity in correlations [33], it has become a routine to combine objective weighting with ranking techniques in sustainable supply chain management [34, 35], including the analysis of sustainability criteria under uncertainty [36].

For ranking purposes, fuzzy TOPSIS compares alternatives based on their proximity to the positive ideal and distance from the negative ideal [37]. Expert judgments expressed in language form are converted into triangular fuzzy numbers [38], thus keeping the element of uncertainty in the evaluation process intact [39]. The use of fuzzy TOPSIS has found widespread application in sustainable supply chain management and supplier selection [40], including works conducted in the Jordanian garments [41], Turkish textiles [42], and Bangladeshi ready-made garments [43] industries. On the other hand, grey relational analysis is selected as an additional method of ranking due to its ability to handle limited and incomplete data [44]. In contrast to fuzzy TOPSIS, grey relational analysis uses reference sequence similarity as the base for comparison [45], and the method has been successfully combined with fuzzy TOPSIS for sustainable supplier evaluation [46]. Therefore, instead of depending upon one ranking system, the agreement between the two approaches increases confidence in the findings, while any difference shows potentially unstable rankings [47].

2.4 Supply Chain Finance Measurements

Twenty-one sub-criteria were generated from the supply chain finance, sustainable supply chain, and sustainable supply chain finance literature and divided into six criteria. Based on the concept of the triple bottom line, the social, environmental, and economic aspects were further expanded to include operational capacity, product and service management, and organizational policies. This expands not only the outcomes of sustainability but also the organizational capabilities for implementing supply chain finance successfully. The sub-criteria and their sources are presented in Table 1. The social factors refer to the relationships that support supply chain finance. Stakeholder engagement (C1) involves stakeholder participation in supply chain finance decisions [48], whereas stakeholder and customer satisfaction (C2) contributes to competitive performance [11]. The buyer-supplier relationship (C3) involves relationship quality and affects both performance and financial results [49], especially where buyer-financed supply chain finance is concerned.

Environmental factors include both organizational commitment to the environment and implementation. Environmental policies and practices (C4) help achieve resource efficiency and competitive advantage [48]; reduction, reuse, and recycling of wastewater and energy (C5) ensure that these commitments are translated into actions, which have financial, social, and environmental consequences [50]. The economic factors represent the working capital factors that have a direct bearing on supply chain finance. Inventory management (C6), acquisition of raw materials (C7), and service delivery management (C8) affect costs, delivery, and funding needs [3]. Price and cost information (C9) provides financial visibility necessary for evaluating the costs and benefits of sustainable supply chain finance [51]. Operational capacity factors are an indication of the efficiency with which a company can handle its operations and supply chain management. Demand forecasting (C10), resource management (C11), and inventory efficiency (C12) help in efficient planning [12], while inventory efficiency is all about making efficient use of inventory [3]. Mismatch between supply chain strategy and business strategy (C13) is another factor that matters, as poor alignment affects efficiency [52].

The product and service management factors include customer service availability (C14), product renovation for customers (C15), and quality of products and services (C16). This demonstrates the firm's capability to fulfill the changing demands of customers and the market [12, 51]. Unavailability of third-party e-platforms (C17) becomes more significant since digital platforms enable supply chain finance activities and supply chain coordination [4, 6]. Organizational policies can enable and facilitate supply chain finance adoption. Lack of common vision (C18), lack of collaboration (C19), and lack of communication (C20) may prevent coordination and information exchange among supply chain finance stakeholders [2]. Likewise, inadequate government support (C21) can hamper the growth of sustainable supply chain finance by way of regulatory and institutional constraints, especially in developing

countries [53]. Throughout this study, the six higher-level elements are referred to as main criteria, and the 21 elements nested within them as sub-criteria.

Table 1. Hierarchy of the main criteria and sub-criteria

Criteria	No.	Sub-Criteria	Sources
Social factors	C1	Stakeholder engagement	[48]
	C2	Stakeholder and customer satisfaction	[11]
	C3	Buyer–supplier partnership level	[49]
Environmental factors	C4	Environmental policies and practices	[48]
	C5	Reduction, reuse, and recycling of wastewater and energy	[50]
	C6	Inventory control	[3]
Economic factors	C7	Raw material procurement	[3]
	C8	Service delivery management policies	[3]
	C9	Price and cost information	[51]
Operational capacity	C10	Demand management and forecasting	[12]
	C11	Resource management	[12]
	C12	Inventory control and efficiency	[3]
	C13	Mismatch between supply chain strategy and business strategy	[52]
Product and service management	C14	Customer service availability	[51]
	C15	Product renovation covering customer segments	[12]
	C16	Product and service quality	[51]
	C17	Unavailability of third-party e-platforms	[4]
Organizational policies	C18	Absence of a shared vision among supply chain partners	[2]
	C19	Weak collaboration among partners	[2]
	C20	Poor communication between partners	[2]
	C21	Absence of government support	[53]

2.5 Research Gaps and Highlights

Despite the growing literature on supply chain finance and sustainability, several important gaps remain. First, existing supply chain finance readiness frameworks primarily emphasize financial and economic factors, while environmental, social, operational, technological, and organizational dimensions receive comparatively limited attention [7]. The traditional triple-bottom-line framework identifies desired sustainability outcomes but does not sufficiently capture the organizational capacities, policies, and product and service management practices needed to implement sustainable supply chain finance successfully. Second, most supply chain finance studies have been conducted in developed economies with mature financial institutions, reliable credit-rating systems, legally transferable receivables, and well-established digital infrastructure. These institutional assumptions may not apply to emerging economies, where access to external finance is limited and supply chain finance adoption may depend more strongly on buyer–supplier relationships, stakeholder cooperation, trust, government support, and digital readiness [8, 9, 22]. Therefore, the relevance and relative importance of existing sustainable supply chain finance criteria require further examination within an emerging-economy context.

Third, previous studies concerning sustainable supply chain finance assessment have predominantly employed subjective weighting methods, such as the analytic hierarchy process and the best-worst method. Although these methods effectively capture expert preferences, their results may be influenced by respondent bias and judgment inconsistency, particularly when numerous criteria are considered [13]. The application of objective weighting methods that account for both criterion variability and intercriteria relationships remains limited in the assessment of sustainable supply chain finance readiness. Finally, many previous studies have relied on a single ranking method, limiting the ability to evaluate the stability and robustness of the resulting priorities. Few studies have combined objective weighting with multiple ranking techniques capable of addressing uncertain, incomplete, and linguistically expressed industry assessments.

To address these gaps, this study develops a multidimensional sustainable supply chain finance readiness framework comprising six criteria and 21 sub-criteria for the manufacturing industry of an emerging economy. It integrates CRITIC with fuzzy TOPSIS and grey relational analysis to derive data-driven weights, compare rankings, and identify priority areas for improvement. The main contributions are as follows:

- It extends the triple bottom line by incorporating operational capacity, product and service management, and organizational policies, enabling a more holistic assessment of sustainability and financial readiness.
- It uses CRITIC to derive weights for 21 sub-criteria from assessment data, reducing reliance on subjective pairwise comparisons.
- It combines fuzzy TOPSIS and the grey relational analysis to provide complementary rankings and strengthen the robustness of the findings.

- It identifies priority areas for improving sustainable supply chain finance readiness and provides guidance for manufacturers, financial institutions, policymakers, and other stakeholders in emerging economies.

3 Proposed Framework

This study proposes an integrated multi-criteria decision-making framework for evaluating sustainable supply chain finance readiness under uncertainty. The six sustainable supply chain finance criteria and 21 sub-criteria were identified through a literature review and evaluated by two separate groups serving distinct analytical purposes. First, 11 respondents from manufacturing organizations and financial institutions in Bangladesh assessed the importance of the sub-criteria using the linguistic scale defined in Section 3.1. The respondents held supply chain, procurement, or finance roles, including officers from banks and non-bank financial institutions engaged in trade financing, and were selected purposively based on at least ten years of direct involvement in supply chain or trade finance decisions. The resulting 11×21 response matrix was normalized, and CRITIC was applied to derive data-driven weights based on sub-criterion variability and intercriteria correlations. Second, three experts with at least 15 years of combined experience in supply chain management and trade finance evaluated the six criteria against the 21 sub-criteria using the scales defined in Sections 3.2 and 3.3. A small panel is appropriate for this task, which requires cross-dimensional judgment rather than statistical representativeness. Sub-criteria C13, C17, C18, C19, C20, and C21 were treated as non-beneficial because they represent barriers. Experts rated the significance of each sub-criterion with respect to each criterion. For enabling sub-criteria, a higher rating indicates a stronger contribution to readiness; for barrier sub-criteria, a higher rating indicates a more severe barrier. The barrier sub-criteria were therefore normalized as smaller-the-better. The fuzzy assessments were converted into triangular fuzzy numbers, and the individual judgments were aggregated. Fuzzy TOPSIS ranked the six criteria according to their distances from the positive and negative ideal solutions, while the grey relational analysis ranked them based on their similarity to a reference sequence. Applying both methods to the same CRITIC-weighted inputs enables cross-method comparison and evaluation of ranking stability [35, 36]. The proposed framework is illustrated in Figure 1, and its methods are detailed in the following subsections.

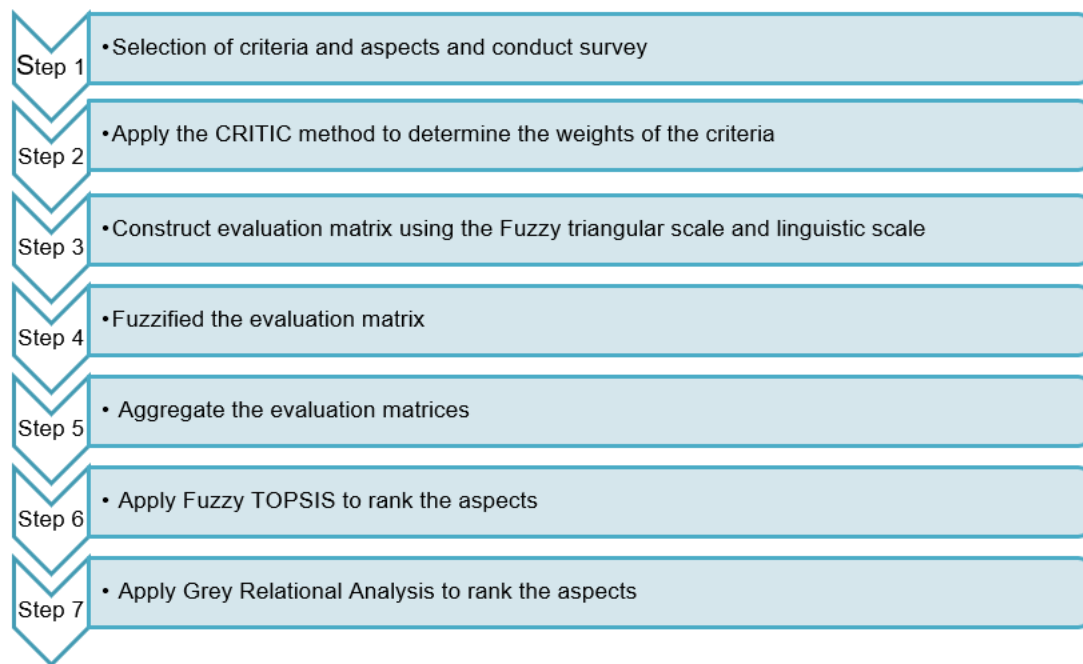


Figure 1. Steps in the proposed framework

Note: TOPSIS = Technique for Order Preference by Similarity to Ideal Solution; CRITIC = Criteria Importance Through Intercriteria Correlation.

3.1 Criteria Importance Through Intercriteria Correlation

The category of correlation-based methods includes approaches such as the CRITIC method [14]. The CRITIC method aims to analyze the decision matrix to determine the amount of information provided by each criterion. In contrast to subjective approaches where preferences of the decision maker must be taken into consideration, CRITIC does not use any preferences because weights are derived from the dispersion of the criterion and from the correlation

between criteria. Therefore, criteria that provide highly redundant information are assigned lower weights [33]. Let x_{ij} denote the rating given by the 11 respondents $i(i = 1, \dots, m)$ to 21 criteria $j(j = 1, \dots, n)$, using the scale presented in Table 2.

Table 2. Linguistic scale for assessing criterion significance in the Criteria Importance Through Intercriteria Correlation (CRITIC) method [11, 12]

Linguistic Term	Numerical Value
Very low	0.2
Low	0.4
Moderate	0.6
High	0.8
Very high	1.0

Step 1: Each of the criteria x_{ij} has its own membership function r_{ij} , which maps all values of the criteria f_j to the interval [0,1]. This mapping is carried out according to the idea of the ideal point. Beneficial criteria are mapped such that their maximum values correspond to the ideal point, whereas non-beneficial criteria are mapped such that their minimum values correspond to the ideal point. As a result, the initial matrix becomes a matrix with elements r_{ij} .

$$r_{ij} = \begin{cases} \frac{x_{ij} - x_j^{\min}}{x_j^{\max} - x_j^{\min}}, & \text{for beneficial criteria} \\ \frac{x_j^{\max} - x_{ij}}{x_j^{\max} - x_j^{\min}}, & \text{for non-beneficial criteria} \end{cases} \quad (1)$$

Step 2: Each vector has its own standard deviation, which shows the level of divergence of the variant values according to a certain criterion from its mean value, and each pair of criteria has its own correlation coefficient, which indicates the level of their conveying the same information. The standard deviation of criterion j is $\sigma_j = \sqrt{\sum_{i=1}^m (r_{ij} - \bar{r}_j)^2 / (m - 1)}$, where $\bar{r}_j$ is the mean of the normalized column j , and r_{jk} is the Pearson linear correlation coefficient between the normalized columns of criteria j and k . The amount of information C_j contained in the criteria j is determined in the following manner:

$$C_j = \sigma_j \sum_{k=1}^n (1 - r_{jk}) \quad (2)$$

Step 3: Sub-criterion weights are obtained by normalizing the values C_j so that they sum to unity:

$$w_j = \frac{C_j}{\sum_{k=1}^n C_k}, \sum_{j=1}^n w_j = 1 \quad (3)$$

3.2 Fuzzy Technique for Order Preference by Similarity to Ideal Solution

Fuzzy TOPSIS is employed in this study to analyze the performance of the aspects of supply chain finance and assign a priority score accordingly. The chosen alternative must have the smallest distance from the positive-ideal solution and the largest distance from the negative-ideal solution, as per the methodology [37]. The closeness to the positive-ideal solution and the distance from the negative-ideal solution are two measures defined by TOPSIS. This method then selects the alternative that is close to the ideal solution [54]. As the judgments collected in this case are linguistic, each is represented as a triangular fuzzy number $\tilde{a} = (a_1, a_2, a_3)$, which preserves the imprecision of expert language through the calculation [38, 55, 56].

Step 1: Obtain the subjective judgments of the decision-makers regarding the importance of the criteria, using the fuzzy triangular scale presented in Table 3.

Step 2: Calculate the fuzzy significance of the criteria based on the decision makers' subjective evaluations by aggregating the judgments of the $K = 3$ experts, taking the smallest lower bound, the mean modal value and the largest upper bound so that the full range of expert disagreement is retained.

$$a_{ij} = \min_k a_{ij}^k, b_{ij} = \frac{1}{K} \sum_{k=1}^K b_{ij}^k, c_{ij} = \max_k c_{ij}^k \quad (4)$$

Step 3: Normalize the fuzzy decision matrix. After calculating $c_j^* = \max_i c_{ij}$ for the beneficial criteria by taking the maximum value and $a_j^- = \min_i a_{ij}$ for non-beneficial criteria by taking the minimum value, the normalization

process can be performed by the following formula, in which the reciprocal form reverses the order of the fuzzy components so that all normalized values remain within [0,1].

$$\tilde{r}_{ij} = \begin{cases} \left(\frac{a_{ij}}{c_j^*}, \frac{b_{ij}}{c_j^*}, \frac{c_{ij}}{c_j^*} \right), & \text{for beneficial criteria} \\ \left(\frac{a_j}{c_{ij}}, \frac{a_j}{b_{ij}}, \frac{a_j}{a_{ij}} \right), & \text{for non-beneficial criteria} \end{cases} \quad (5)$$

Step 4: Calculate the weighted normalized decision matrix with this formula, where w_j is the weight of criterion j obtained from the CRITIC method in Eq. (3):

$$\tilde{v}_{ij} = w_j \cdot \tilde{r}_{ij} \quad (6)$$

Step 5: Compute the fuzzy positive ideal solution and the fuzzy negative ideal solution. The fuzzy positive ideal solution and fuzzy negative ideal solution of the alternatives are computed as follows, where $\tilde{v}_j^*$ represents the fuzzy positive ideal value and $\tilde{v}_j^-$ is the fuzzy negative ideal value for criterion j :

$$A^* = \{\tilde{v}_1^*, \tilde{v}_2^*, \dots, \tilde{v}_n^*\}, \tilde{v}_j^* = \max_i \tilde{v}_{ij}; A^- = \{\tilde{v}_1^-, \tilde{v}_2^-, \dots, \tilde{v}_n^-\}, \tilde{v}_j^- = \min_i \tilde{v}_{ij} \quad (7)$$

Step 6: Calculate the distance of each alternative from A^* to A^- . The distance between two triangular fuzzy numbers is measured by the vertex method, $d(\tilde{a}, \tilde{b}) = \sqrt{1/3 [(a_1 - b_1)^2 + (a_2 - b_2)^2 + (a_3 - b_3)^2]}$, and the separation of each alternative is the sum of its criterion-wise distances:

$$d_i^+ = \sum_{j=1}^n d(\tilde{v}_{ij}, \tilde{v}_j^*), i = 1, \dots, m \quad (8)$$

$$d_i^- = \sum_{j=1}^n d(\tilde{v}_{ij}, \tilde{v}_j^-), i = 1, \dots, m \quad (9)$$

where, d_i^+ denotes the distance between each alternative and the ideal solution, and d_i^- denotes the distance between each alternative and the negative-ideal solution.

Step 7: Calculate the closeness coefficient. The closeness coefficient of alternative A_i with respect to the ideal solution A^* can be defined as:

$$CC_i = \frac{d_i^-}{d_i^+ + d_i^-}, 0 \leq CC_i \leq 1, i = 1, \dots, m \quad (10)$$

Clearly, an alternative A_i is closer to A^* than to A^- as CC_i approaches 1, suggesting that the evaluation grade of A_i increases with CC_i . The closeness coefficient CC_i can be regarded as the evaluation value of alternative A_i . Thus, the larger CC_i , the higher priority the alternative A_i .

Table 3. Triangular fuzzy scale for assessing the significance of sub-criteria in fuzzy TOPSIS [38, 39, 55]

Linguistic Significance Level	Triangular Fuzzy Number
Extremely low significance	(1, 1, 2)
Very low significance	(1, 2, 3)
Low significance	(2, 3, 4)
Medium significance	(3, 4, 5)
High significance	(4, 5, 6)
Very high significance	(5, 6, 7)
Extremely high significance	(6, 7, 7)

Note: TOPSIS = Technique for Order Preference by Similarity to Ideal Solution.

3.3 Grey Relational Analysis

Proposed by Deng [57], the grey relational analysis is the most popular model of the grey system. The application of the model lies in the determination of an optimal value of the process parameters, and it is frequently used for assessing the performance of a complicated project with insufficient data [58]. By assigning weights to different individuals' responses, grey relational analysis can be used to determine the optimal aspects for solving problems with multiple criteria [44, 45]. Let $x_i^0(k)$ denote the aggregated rating of alternative i on criterion k using Table 4.

Table 4. Linguistic scale for assessing the significance of sub-criteria in grey relational analysis [44]

Linguistic Rating	Numerical Score
Extremely low significance	1
Very low significance	2
Low significance	3
Medium significance	4
High significance	5
Very high significance	6
Extremely high significance	7

Step 1: Data pre-processing and normalizing. The data to be used in grey analysis must be preprocessed into quantitative indices for normalizing raw data for further analysis. Preprocessing raw data is a process of converting an original sequence into a decimal sequence between 0.00 and 1.00 for comparison. If the expected data sequence is of the “higher-the-better” form, then the original sequence can be normalized as:

$$x_i^*(k) = \frac{x_i^0(k) - \min_i x_i^0(k)}{\max_i x_i^0(k) - \min_i x_i^0(k)} \quad (11)$$

where, $x_i^0(k)$ is the original sequence, $x_i^*(k)$ is the sequence after the data preprocessing, $\max_i x_i^0(k)$ is the largest value of $x_i^0(k)$, and $\min_i x_i^0(k)$ is the smallest value of $x_i^0(k)$. When the “smaller-the-better” form becomes the expected value of the data sequence, which applies to the six barrier criteria identified above, the original sequence can be normalized as:

$$x_i^*(k) = \frac{\max_i x_i^0(k) - x_i^0(k)}{\max_i x_i^0(k) - \min_i x_i^0(k)} \quad (12)$$

Step 2: Deviation sequence. The deviation sequence of the reference sequence $x_0^*(k) = 1$ is given by:

$$\Delta_{0i}(k) = |x_0^*(k) - x_i^*(k)|, \Delta_{\max} = \max_i \max_k \Delta_{0i}(k), \Delta_{\min} = \min_i \min_k \Delta_{0i}(k) \quad (13)$$

Step 3: Grey relational coefficient. The grey relational coefficient is calculated to express the relationship between the ideal and actual normalized results. Thus, the grey relational coefficient can be expressed as:

$$\xi_i(k) = \frac{\Delta_{\min} + \zeta \Delta_{\max}}{\Delta_{0i}(k) + \zeta \Delta_{\max}} \quad (14)$$

where, $\Delta_{0i}(k)$ is the deviation sequence of the reference sequence, and ζ is the distinguishing or identification coefficient, with $\zeta \in [0, 1]$. The value $\zeta = 0.5$ is generally used and is adopted in this study. It follows from Eq. (14) that $\xi_i(k)$ is bounded below by $\zeta/(1 + \zeta)$ and above by 1.

Step 4: Grey relational grade. After obtaining the grey relational coefficient, the grey relational grade is obtained by combining the coefficients across criteria. Because the criterion weights w_k derived in Eq. (3) sum to unity, the grade is computed as their weighted sum, so that the objective weights inform the ranking in the same way as in fuzzy TOPSIS. The grey relational grade is defined as:

$$\gamma_i = \sum_{k=1}^n w_k \xi_i(k) \quad (15)$$

The grade γ_i therefore inherits the bounds of the coefficient, $\gamma_i \in [\zeta/(1 + \zeta), 1]$, which for $\zeta = 0.5$ gives $[0.333, 1]$. A higher grade indicates greater similarity to the reference sequence and therefore higher priority.

4 Results and Discussion

This section reports the sub-criterion weights obtained from CRITIC, the criteria rankings produced by fuzzy TOPSIS and the grey relational analysis, the agreement between the two methods, and the interpretation of these results.

4.1 Sub-Criterion Weights

The CRITIC method was used for the 11×21 matrix given in Section 3 using Eqs. (1)–(3) and the results are shown in Table 5. The sub-criterion weights add up to 1 and vary between 0.030 and 0.068, whereas the benchmark is $1/21 = 0.048$.

The six most heavily weighted sub-criteria are the mismatch between supply chain and business strategies (C13, 0.068), raw material procurement (C7, 0.064), unavailability of third-party e-platforms (C17, 0.062), stakeholder and customer satisfaction (C2, 0.059), environmental policies and practices (C4, 0.057), and customer service availability (C14, 0.054). The four least heavily weighted are weak collaboration among partners (C19, 0.030), reduction, reuse and recycling of wastewater and energy (C5, 0.038), stakeholder engagement (C1, 0.040), and price and cost information (C9, 0.040). This should be carefully considered. In the case of CRITIC, weights are derived from the intensity of contrast and correlation between criteria, but not preferences. Thus, a weight reveals a point at which respondents had the sharpest discrimination and where a sub-criterion had some information that was not conveyed by other sub-criteria. Therefore, the low weight of a sub-criterion should not be interpreted as its lack of importance. On the contrary, it shows that the respondents answered the question in the same way or according to the way their answers were predicted. The small spread of the weight values has its own significance. The largest weight value is 2.2 times the lowest one, and the average deviation from the equal weights model is 0.007. In other words, the weights used are only mildly discriminating, which implies that the rankings presented below are largely dependent on the expert evaluation matrix.

Table 5. Weights of the sub-criteria obtained from the CRITIC

Criterion	Weight	Criterion	Weight
C1	0.040	C12	0.043
C2	0.059	C13	0.068
C3	0.045	C14	0.054
C4	0.057	C15	0.041
C5	0.038	C16	0.043
C6	0.051	C17	0.062
C7	0.064	C18	0.044
C8	0.041	C19	0.030
C9	0.040	C20	0.050
C10	0.042	C21	0.045
C11	0.044		

Note: Weights are rounded to three decimal places; the unrounded weights sum to 1 and were used in all subsequent calculations; CRITIC = Criteria Importance Through Intercriteria Correlation.

4.2 Main Criteria Ranking Using the Fuzzy Technique for Order Preference by Similarity to Ideal Solution

The six criteria were assessed on all 21 sub-criteria by a panel of three experts. Their opinions were represented as triangular fuzzy numbers using Eq. (4) and normalized using Eq. (5). The normalized fuzzy decision matrix was then weighted using the CRITIC weights in Table 5 according to Eq. (6). The fuzzy positive ideal solution and fuzzy negative ideal solution were determined using Eq. (7). The corresponding separation distances were calculated using Eqs. (8) and (9), and the closeness coefficient was calculated using Eq. (10). The resulting fuzzy TOPSIS rankings are presented in Table 6.

Table 6. Ranking of the six criteria using the fuzzy TOPSIS

Criteria	No.	d_i^+	d_i^-	CC_i	Rank
Social factors	Criteria 1	0.200	0.261	0.566	1
Operational capacity	Criteria 4	0.213	0.255	0.545	2
Economic factors	Criteria 3	0.229	0.235	0.506	3
Product and service management	Criteria 5	0.267	0.195	0.421	4
Environmental factors	Criteria 2	0.302	0.162	0.350	5
Organizational policies	Criteria 6	0.356	0.110	0.236	6

Note: fuzzy TOPSIS = fuzzy Technique for Order Preference by Similarity to Ideal Solution.

Social factors rank first, followed by operational capacity, economic factors, product and service management, environmental factors, and organizational policies. In addition to the ranking order, the distribution of the closeness

coefficients provides further insight into the relative positions of the criteria. In the first three criteria, there is an overall difference of 0.060, while there is only a difference of 0.021 between the first- and second-ranked criteria. However, the difference between the third- and sixth-ranked criteria is 0.270. These results suggest a relatively clear distinction between the three highest-ranked criteria and the lower-ranked criteria, with organizational policies occupying a particularly weak position relative to the leading criteria.

4.3 Main Criteria Ranking Using the Grey Relational Analysis

The same expert judgments were converted to crisp values, normalized with Eqs. (11) and (12) and used to compute deviation sequences and grey relational coefficients with Eqs. (13) and (14) at a distinguishing coefficient of $\zeta = 0.5$. The grey relational grade was obtained as the CRITIC-weighted sum of coefficients using Eq. (15), as shown in Table 7.

Table 7. Ranking of the six criteria using the grey relational analysis

Criteria	No.	Grey Relational Grade (γ_i)	Rank
Social factors	Criteria 1	0.695	1
Economic factors	Criteria 3	0.645	2
Operational capacity	Criteria 4	0.619	3
Product and service management	Criteria 5	0.601	4
Environmental factors	Criteria 2	0.556	5
Organizational policies	Criteria 6	0.454	6

Social factors again rank first and organizational policies rank last. Because the criterion weights sum to unity, each grade lies within the theoretical interval $[\zeta/(1+\zeta), 1] = [0.333, 1]$, as required. The ranking obtained using grey relational analysis is broadly consistent with that obtained using fuzzy TOPSIS. The difference between the first- and third-ranked criteria is only 0.076. In contrast, organizational policies, with a grey relational grade of 0.454, are 0.102 below the fifth-ranked environmental factors. The two techniques therefore yield the same highest- and lowest-ranked criteria, with a single reversal between the second and third positions.

4.4 Comparison of the Two Methods

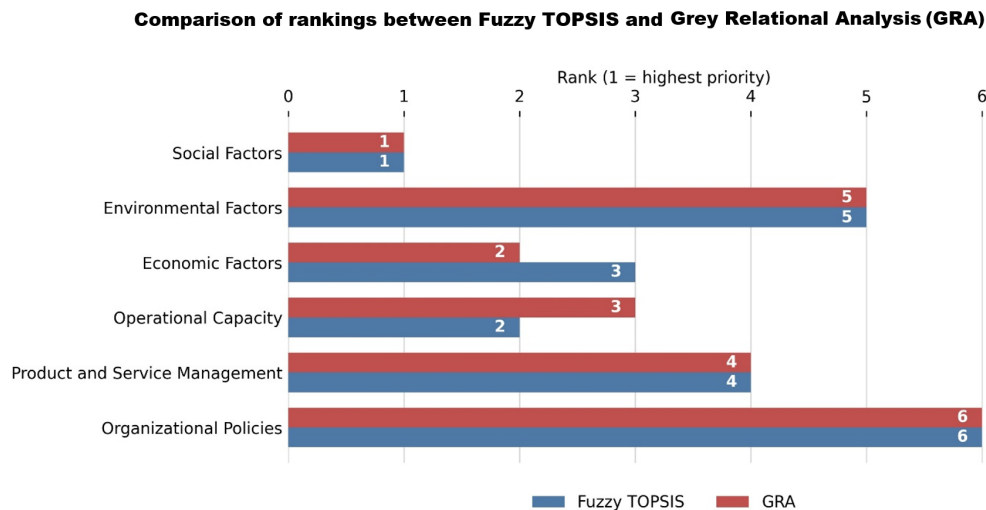


Figure 2. Ranking comparison between the fuzzy TOPSIS and grey relational analysis

Note: fuzzy TOPSIS = fuzzy Technique for Order Preference by Similarity to Ideal Solution.

The ranking methods are highly correlated but not completely identical. The only discrepancy is that while fuzzy TOPSIS assigns operational capacity as the second criterion and economic factors as the third one, grey relational analysis ranks them vice versa. The remaining criteria have the same ranking for both methods. The discrepancy arises from the reasoning behind the aggregation process in both techniques, not from the inconsistency of the dataset. The fuzzy TOPSIS technique compares each of the alternatives against both the positive and negative ideal solutions, so that the aspect is penalized for closeness to the lowest value seen in the observations and rewarded for closeness to the highest value. The grey relational analysis technique only compares each of the alternatives against one

reference sequence and thus does not take into consideration the negative-ideal side. Operational capacity performs better relative to the negative ideal, which contributes to its higher ranking under the fuzzy TOPSIS technique, whereas economic factors show greater similarity to the reference sequence. The combined procedure provides more information than each procedure individually. This agreement indicates that the highest and lowest rankings are not artefacts of a particular aggregation method, while the disagreement identifies the specific pair of adjacent criteria whose relative ordering is sensitive to the aggregation method. Figure 2 presents the comparison between the two rankings.

4.5 Discussion

The findings consistently show that the criterion of organizational policies, with the lowest ranking in both fuzzy TOPSIS (0.236) and grey relational analysis (0.454), requires the greatest improvement. This criterion involves shared vision, collaboration, communication between the members of the supply chain, and government policies. The distinctiveness of this criterion indicates that the weakness associated with sustainable supply chain finance readiness relates to the inter-organizational and institutional context rather than to the capabilities of individual firms. This is especially true for developing countries where underdeveloped receivable systems increase reliance on buyers. On the sub-criteria level, the mismatch between supply chain strategy and business strategy, raw material sourcing, and Unavailability of third-party e-platforms have relatively higher weights owing to the diversity among organizations in terms of strategic alignment, stable sourcing, and digital readiness. The importance of digital platforms can be attributed to their increasing relevance to the process of financing and supply chain coordination [6]. Nevertheless, CRITIC weights should not be considered as low importance but rather as lower diversity among survey participants. For example, price/cost data is highly important for the assessment of sustainable supply chain finance [51], whereas poor collaboration has the lowest weight within the least important aspect.

Social factors are considered more important than the other two categories using any of the two ranking methods. However, the similarity in social factors, operational capacity and economic factors indicates that the factors fall under one dominant category and not separate rankings. In conclusion, the analysis reveals that whereas the organizations differ in terms of their strategic, procurement and digitization capabilities, they face similar challenges regarding institutional and organizational support. There is therefore a need for improvement at both the organizational and supply chain levels of sustainable supply chain finance readiness.

5 Conclusions and Future Work

This study identified the criteria in sustainable supply chain finance that require the most improvement in the manufacturing industry of emerging economies and the criteria that can be used to differentiate companies according to their preparedness for adopting sustainable supply chain finance. The weighting of 21 sub-criteria under six criteria was performed using the CRITIC method based on responses received from 11 practitioners. Then the criteria were ranked using the fuzzy TOPSIS and grey relational analysis methods by a three-member expert team, with both methods ranking social factors first ($CC_i = 0.566$; $\gamma_i = 0.695$) and organizational policies last ($CC_i = 0.236$; $\gamma_i = 0.454$). The clear separation of organizational policies from the other criteria at the bottom of the ranking can be explained by the composition of this criterion. Partnership vision, cooperation, communication, and government support cannot be seen as competencies that a firm creates for itself but as an environment within which the firm operates. The bottom position of organizational policies indicates that the surveyed companies perceive the surrounding coordination structure as the weakest component, not their own operations. This is consistent with a market where the infrastructure for accounts receivable is poorly developed, and credit is based on a history of relationships rather than formal guarantees; this is why social factors come first, since in such conditions, the buyer's relationship becomes the object against which financing is issued through a contract. The weights of the sub-criteria point in the same direction, but from a different perspective. Companies differ the most in terms of strategic alignment (0.068), raw materials procurement (0.064), and access to third-party e-platforms (0.062).

This study has two practical implications. Organizational-level improvements are unlikely to address the most binding constraint; hence, the important measures would be those related to the legal treatment of receivables assignment, non-bank financing, and common platform infrastructure. From the management perspective, relationship quality cannot be seen as a secondary factor in addition to the financial factors; rather, it is a prerequisite for accessing finance. CRITIC derives the weights from the dispersion and correlation of participants' ratings rather than from explicit pairwise preference comparisons, so the weights are data-driven, although they remain conditional on the underlying linguistic assessments. The resulting hierarchy was further examined using a second ranking method. CRITIC extracts the importance of the factors from the responses of the participants, while the resulting hierarchy is verified using the alternative ranking method.

This study has several limitations that indicate directions for future research. First, the assessment draws on 11 participants and a three-member expert panel. This is sufficient to distinguish the highest-ranked criteria from the lowest-ranked criteria, which is where the principal finding lies, but not to clearly distinguish among the leading

three, whose relative order differs between the two ranking methods. Future work should validate the framework with a substantially larger participant base and an expanded expert panel, which would allow the intercriteria correlations underlying CRITIC to be estimated more precisely and the leading group to be separated with greater confidence. Second, six sub-criteria were framed as barriers and treated as non-beneficial; future studies could restructure them as directional performance constructs so that both specifications can be compared directly. Third, the study covers a single manufacturing context in one emerging economy, so the findings should be tested in other manufacturing sectors, such as pharmaceuticals, electronics, and textiles, and in other emerging economies with comparable financing infrastructure, thereby establishing the extent to which the institutional pattern observed in this study generalizes. Finally, because the rankings rest on expert assessments rather than measured performance, they reflect perceived rather than observed readiness. Future studies could link these priorities to actual financing decisions to test whether the areas identified here predict which firms ultimately obtain supply chain finance.

Author Contributions

Conceptualization, S.M.S.R. and S.M.A.N.; methodology, S.M.S.R. and M.F.T.H.; software, S.M.S.R.; validation, S.M.S.R. and S.M.A.N.; formal analysis, S.M.S.R.; investigation, S.M.S.R. and S.M.A.N.; resources, S.M.S.R.; data curation, S.M.S.R.; writing—original draft preparation, S.M.S.R. and M.F.T.H.; writing—review and editing, S.M.A.N.; visualization, S.M.S.R.; supervision, S.M.A.N. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

Generative AI and AI-assisted technologies were used solely to support language refinement and grammatical editing during the preparation of this manuscript.

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